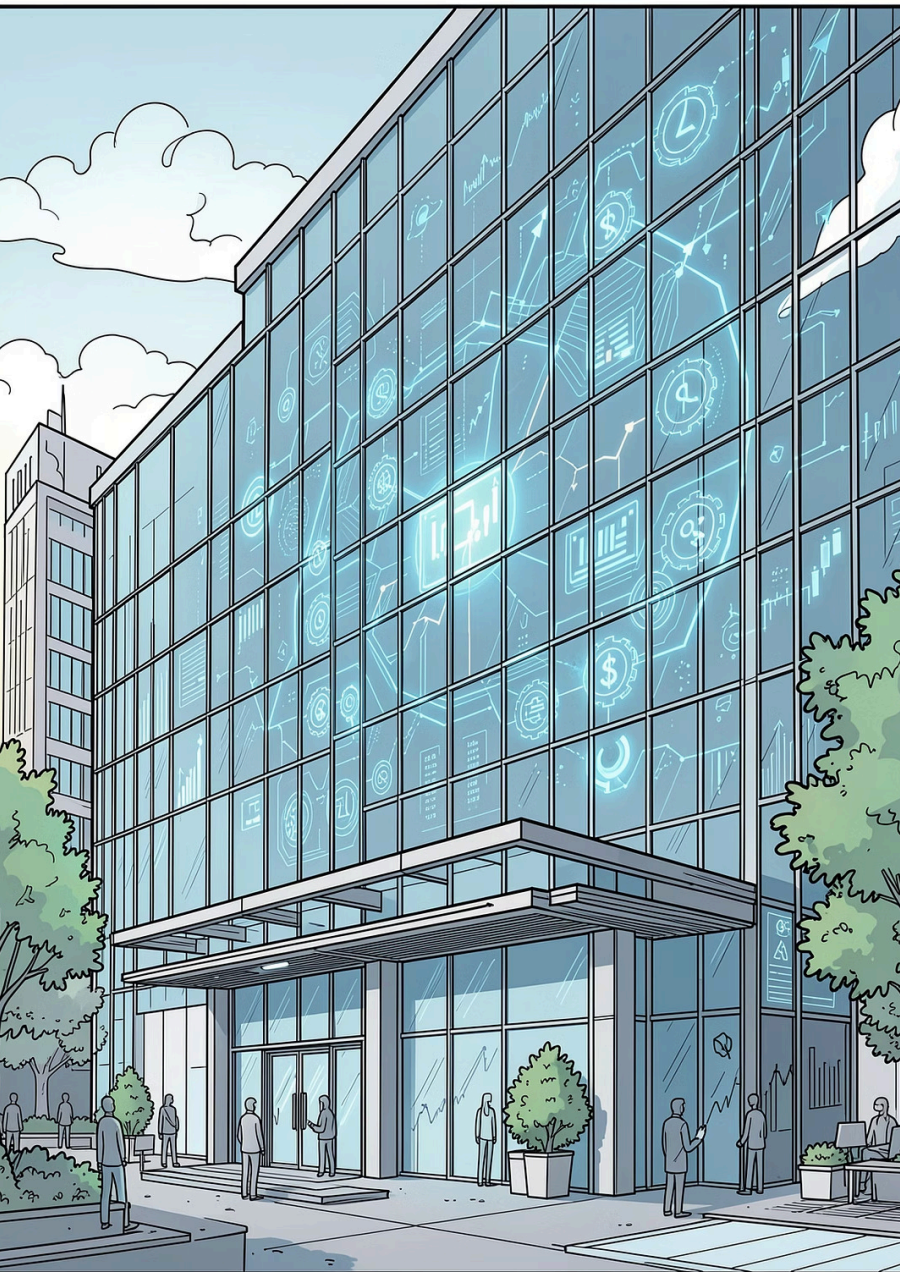




Banking Management System

A full-stack Java application built with Spring Boot, Thymeleaf, and MySQL — delivering secure, real-time banking operations through a clean MVC architecture and responsive UI.

JAVA · SPRING BOOT · MYSQL

Project Objective

This system automates core banking workflows — replacing manual processes with a reliable, digital-first platform. It centralizes customer account management, transaction processing, and data security into a single cohesive application.



→ Automate banking operations

Reduce manual overhead with programmatic transaction handling.

→ Secure data management

Protect customer records with encrypted storage and access controls.

→ Deliver a seamless UI

Provide an intuitive, responsive interface for all user roles.

Core Features at a Glance



Registration & Login

Secure onboarding with credential validation and session management.



Account Management

Create, view, and manage customer accounts with full lifecycle support.



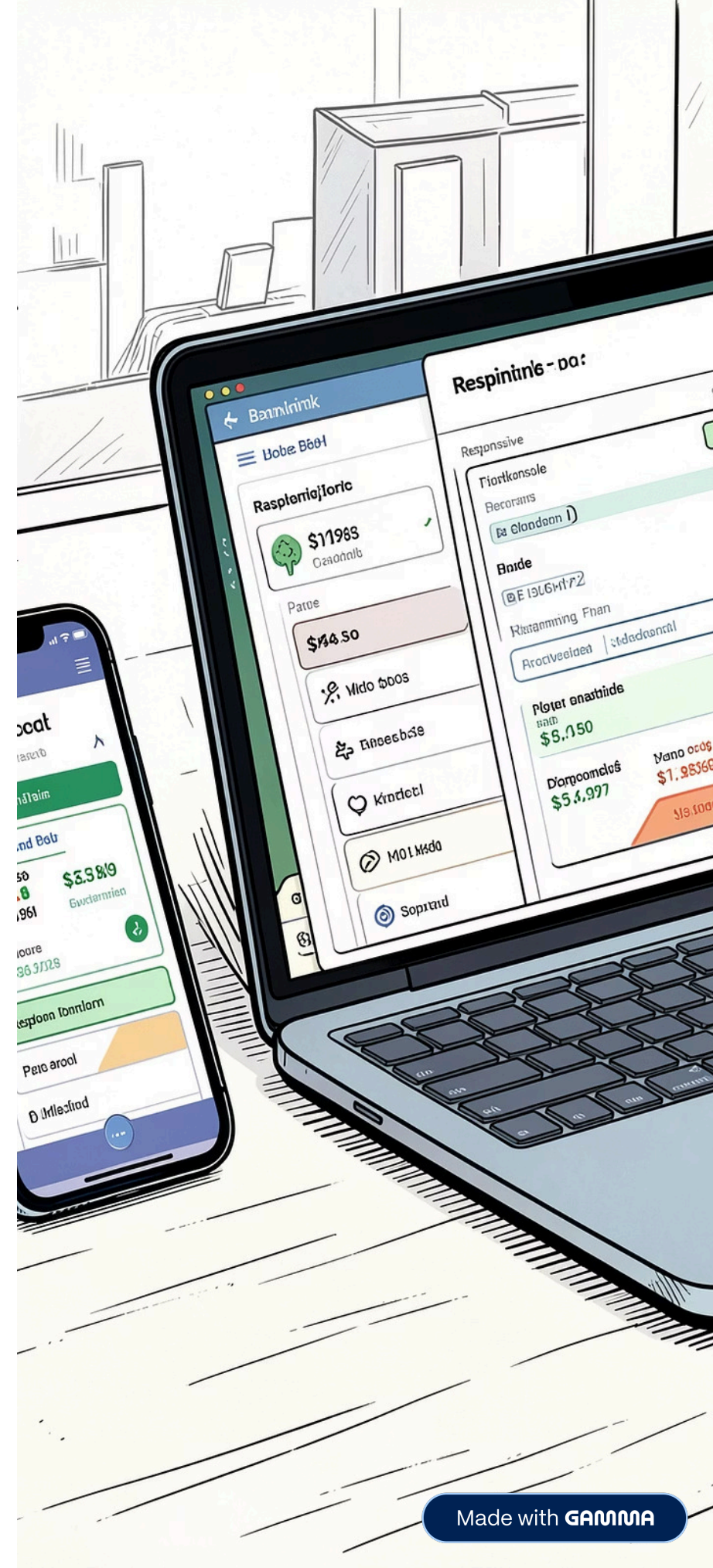
Deposit & Withdrawal

Real-time balance updates with transaction validation and audit trails.

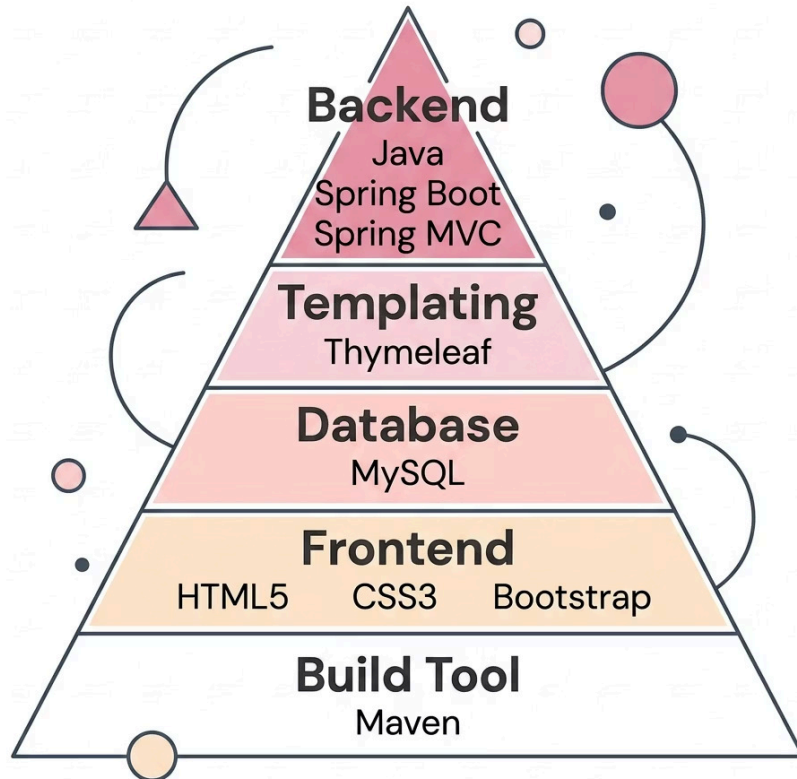


Transaction History

Full chronological record of all account activity with filtering.



Technology Stack



Full-Stack Coverage

Every layer of the application is backed by a mature, industry-standard technology — from server-side Java to responsive Bootstrap frontends.

Backend

Java + Spring Boot + Spring MVC

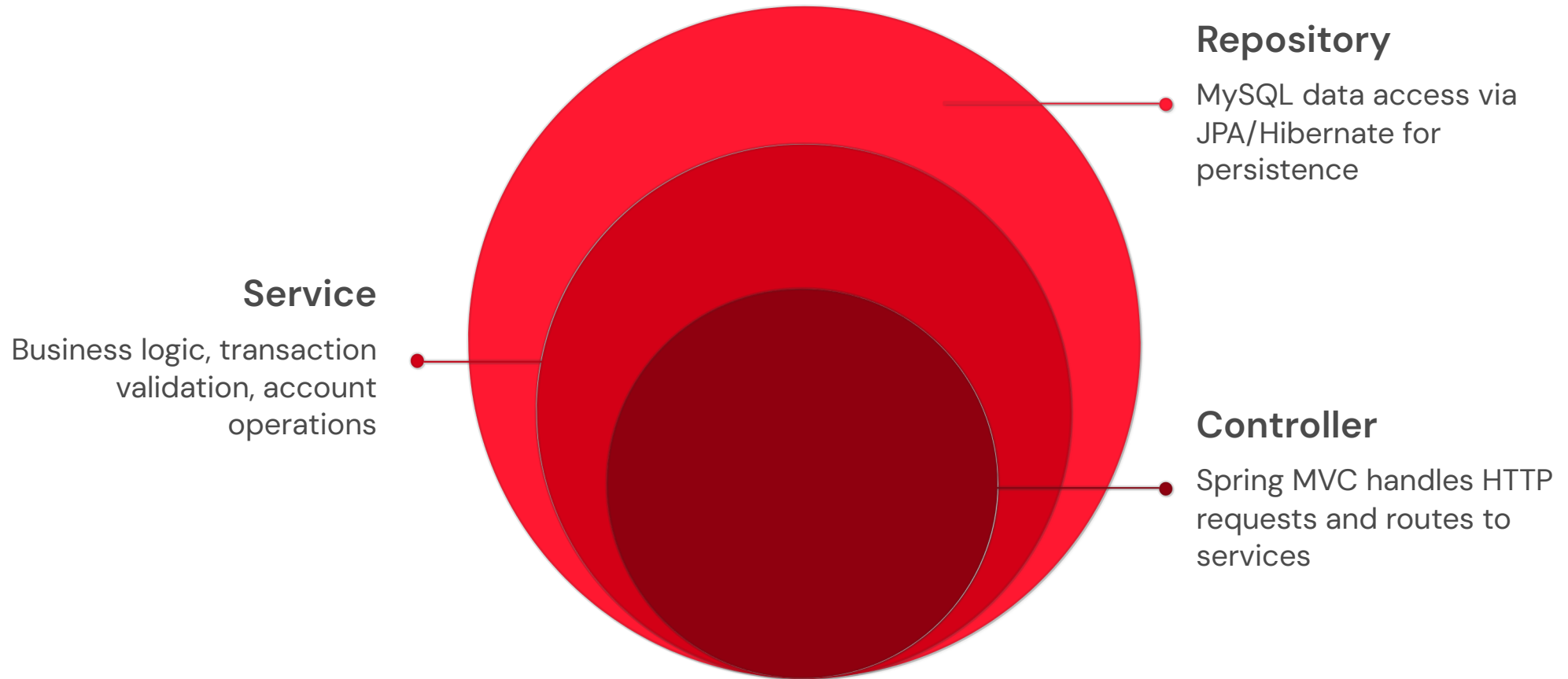
Frontend

Thymeleaf + HTML5 + CSS3 +
Bootstrap

Data Layer

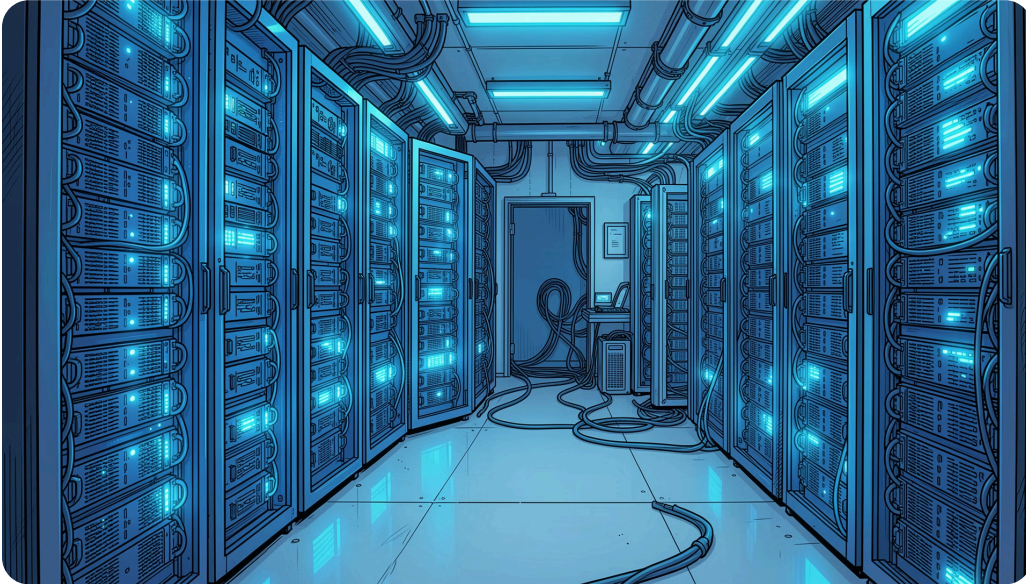
MySQL + Maven

MVC Architecture with Spring Boot



Spring Boot's dependency injection and auto-configuration keep each layer decoupled. Controllers handle web requests, Services enforce banking rules, and Repositories manage persistence — making the codebase testable and maintainable.

Secure Data Storage with MySQL



Relational Data Model

MySQL provides ACID-compliant storage for all banking records. Tables are normalized to eliminate redundancy and enforce referential integrity across customers, accounts, and transactions.

Customer Table

Stores credentials, personal info, and account references.

Account Table

Tracks balance, account type, and status per customer.

Transaction Table

Logs every deposit, withdrawal, and transfer with timestamps.

Responsive UI with Bootstrap



Mobile-First Grid

Bootstrap's 12-column grid adapts layouts seamlessly from desktop to mobile, ensuring accessibility across all device sizes.



Consistent Components

Pre-built Bootstrap components — navbars, cards, modals, and forms — accelerate UI development while maintaining visual consistency.



CSS3 Enhancements

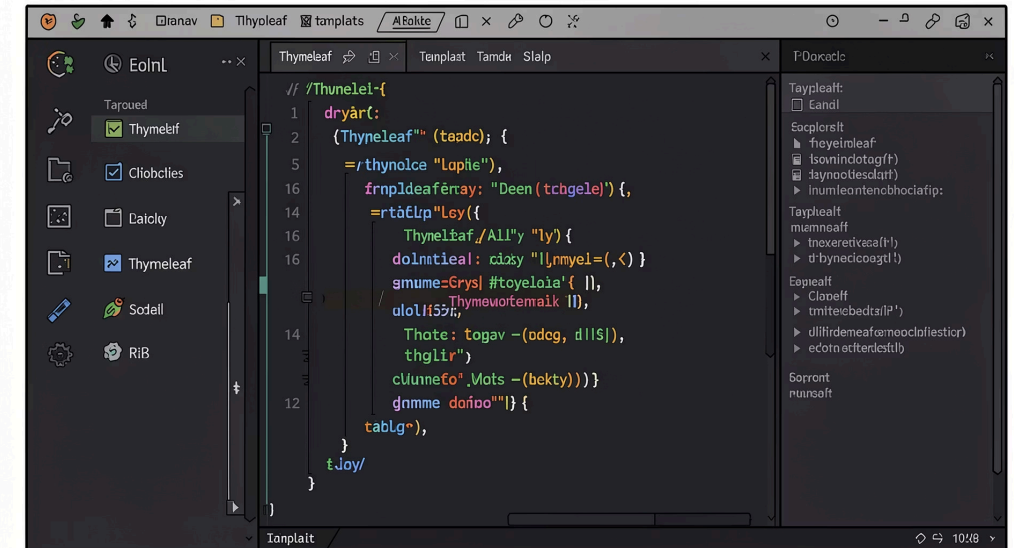
Custom CSS3 builds on Bootstrap's foundation with tailored animations, transitions, and brand-specific styling.



Thymeleaf Server-Side Rendering

Why Thymeleaf?

Thymeleaf is a modern Java template engine that integrates natively with Spring MVC. It renders HTML on the server, reducing client-side complexity and keeping logic centralized.



Spring Integration

Direct binding to Spring model attributes with `th:each`, `th:if`, and `th:text`.

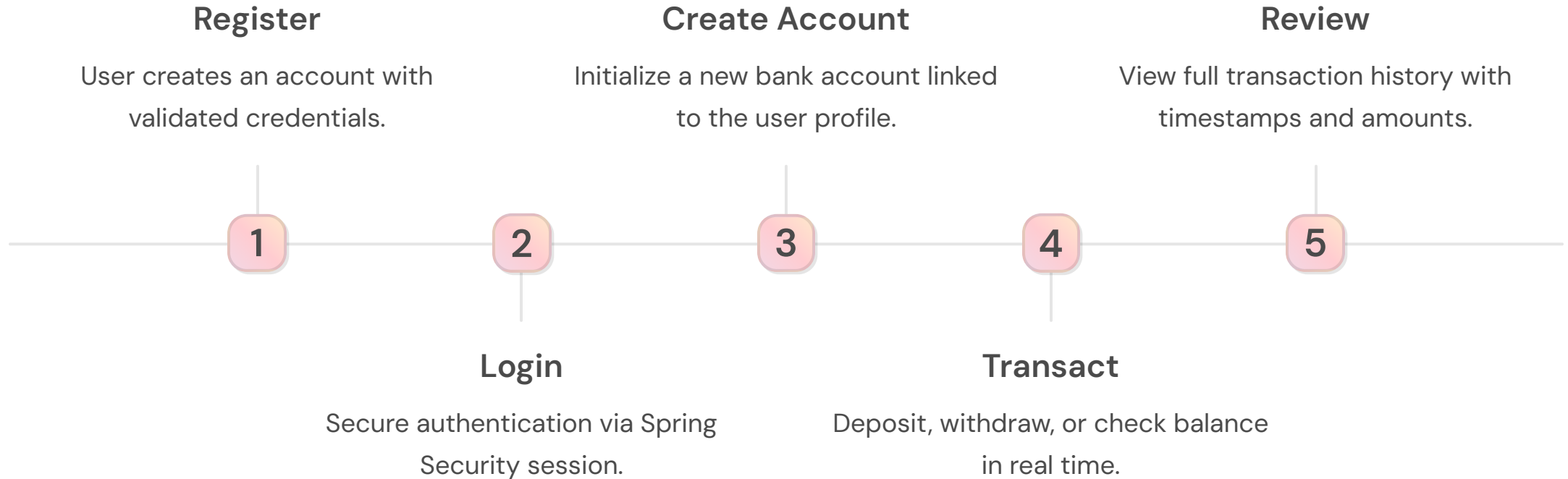
Template Reuse

Fragment-based layouts (`th:replace`) eliminate HTML duplication across views.

Security-Aware

Works with Spring Security to conditionally render UI elements based on user roles.

End-to-End User Flow



i All transactions are persisted to MySQL and immediately reflected in the UI via Thymeleaf model updates — no page reload delays beyond standard request/response.

Key Takeaways

Full Banking Coverage

Registration, accounts, deposits, withdrawals, balance inquiry, and transaction history — all in one system.

Clean MVC Design

Spring Boot + Spring MVC + Thymeleaf separation of concerns keeps the codebase maintainable and testable.

Secure by Design

MySQL ACID compliance, session-based auth, and server-side rendering minimize attack surface.

Responsive & Ready

Bootstrap ensures the interface works across devices — from desktop banking to mobile check-ins.

